*CANDIDATE NUMBER……..*

**THE UNITED REPUBLIC OFTANZANIA**
**PRESIDENT'S OFFICE,**
**REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT**

**FORM TWO MOCK II EXAMINATION**

**SUBJECT CODE 033** **BIOLOGY**
**TIME: 2:30 Hours** **May, 2024**

**INSTRUCTIONS**

1. This paper consists of sections A, B and C with a total of ten (10) questions.
2. Answer all questions in the spaces provided.
3. Section A carries fifteen (15) marks, section B seventy (70) marks and section C carries fifteen (15) marks.
4. All writing must be in blue or black ink except drawing which must be in pencil.
5. Cellular phones and any unauthorized materials are not allowed in the assessment room.

| FOR EXAMINER'S USE ONLY | | |
|---|---|---|
| **QUESTION NO.** | **SCORE** | **MARKER'S INITIAL** |
| **1** | | |
| **2** | | |
| **3** | | |
| | | |
| **7** | | |
| **8** | | |
| **9** | | |
| **10** | | |
| **11** | | |
| **TOTAL** | | |

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**SECTION A (15 Marks)**

(Answer all questions in this section)

1. For each of the items (i) - (x), choose the correct answer from the given alternatives and write its letter in the box provided.

(i) The feeding relationship represented by the sequences of organisms "Green Plants --> Grasshopper --> Frog" is called:

A. Food web
B. Interaction
C. Food chain
D. Pyramid

(ii). Which of the following will be a step to follow in a scientific procedure after formulating a hypothesis?

A. Conclusion
B. Experimentation
C. Observation
D. Problem identification

(iii). Which of the following are communicable diseases?

A. Cancer, gonorrhea and syphilis
B. Cholera, gonorrhea and syphilis
C. Cholera, kwashiorkor and marasmus
D. Cholera, marasmus and rickets

(iv). Which of the following is not a proper method of waste disposal?

A. Use of landfill
B. Make pit latrine
C. Recycling of materials
D. Burning plastic waste at home

(v). An organism with a rigid protective wall made up of chitin material is a representative of which Kingdom?

A. Fungi
B. Monera
C. Plantae
D. Protoctista

(vi) Which one of the following are examples of animal tissues?

A. Muscle, blood and bone
B. Brain, bone and liver
C. Skin, heart and brain
D. Muscle, liver and skin

(vii). Which of the following types of waste cannot be recycled? ..... .

A. paper
B. metal scraps
C. used syringe
D. plastics

(viii). Why are bats and birds not placed in the same species while they are closely related organisms? .....

A. They cannot interbreed to produce a fertile offspring
B. They can interbreed to produce a fertile offspring
C. One lays eggs and another can give birth
D. One possesses feathers and another short beak

(ix). Which one of the following is recommended as a First Aid procedure for a victim who has been stung by bee?

A. Sitting quietly
B. Giving huge amount of water to the victim
C. Applying baking soda paste on the bee stung area.
D. Leaving the cotton on the stung area for few minutes.

(x) What will happen when a red blood cell is placed in a more concentrated solution?

A. Crenation
B. Turgidity
C. Haemolysis
D. Plasmolysis

| i | ii | iii | iv | v | vi | vii | viii | ix | x |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |

2. Match the functions of the digestive enzymes in LIST A with the names of the specific enzymes in LIST B by writing the letter of the correct response below the corresponding item number in a table provided.

| LIST A | LIST B |
|---|---|
| i. It digests Beans into Peptides.<br>ii. It digests Avocado into Fatty acids and Glycerol.<br>iii. It digests Rice into Maltose.<br>iv. It digests Cane sugar into Glucose and Fructose<br>v. It digests Milk sugar into Glucose and Galactose | A. Salivary amylase<br>B. Trypsin<br>C. Maltase<br>D. Sucrase<br>E. Lactase<br>F. Lipase<br>G. Peptidase |

| LIST A | i | ii | iii | iv | v |
|---|---|---|---|---|---|
| LIST B | | | | | |

**SECTION B (70 Marks)**

(Answer All Questions)

3. Fausta and Isaka were in the Laboratory examining cells from different Living things using Light microscope. Fausta saw **Cell A** while Isaka saw **Cell B** as shown in **Figure 1** below. Study their observation and then answer the questions that follow.

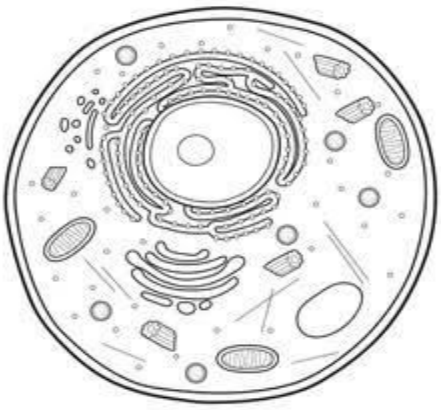
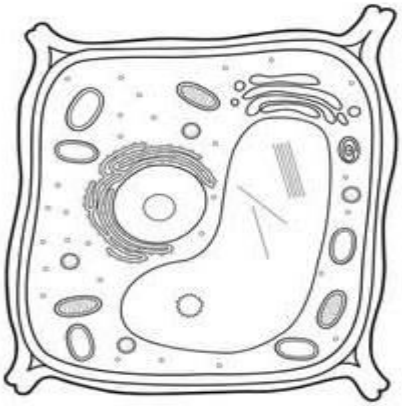
| **Cell A** | **Cell B** |
|---|---|
|  |  |

**Figure 1**

(a) What are the real names of cell A and Cell B?

(b) Tabulate four differences between the two cells named in 3(a).
(c) What are the four similarities between the two cells named in 3(a).

4. Name and draw the Laboratory apparatus that can be used to:
(a) Crush and grind solid complex things into fine powdered or soft simple substances.
(b) Picking up smaller animals such as insects without hurting them.
(c) Estimate the population of living things in an area.
(d) put and observe specimen closely.
(e) Heat, test and store chemicals.
5. Draw a neat diagram of human digestive system and label only regions of digestion in it.
6. **Figure 2** is the diagram of a vector of some communicable diseases. Study it careful and then answer the questions that follow:

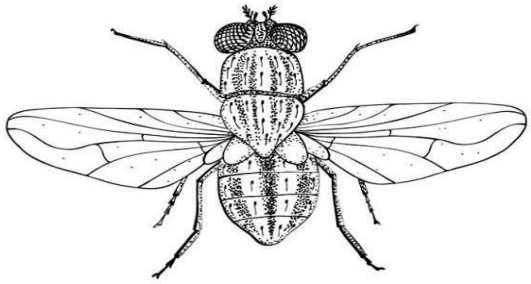


**Figure 2**

(a) What is the common name of the vector in Figure 2 above? _____
(b) Name two common pathogens that can be spread by the vector named in 6(a)
(c) Mention the disease caused by each pathogen named in 6(b)
(d) Explain four common preventive measures of the infections that can be transmitted by the vector named in
7. (a) What does artificial classification and natural classification mean?
(b) What are the similarities between Mushroom and Moss plant? Mention four points.
8. (a) State the functions of the following parts of the microscopes.
(i) Mirror (ii) Arm (iii) Base
(b) Calculate the highest total magnification of the Light microscope with 10x Eyepieces lens and 15x as well as 40x Objective lens magnification.
9. (a) explain four habits that everyone must consider during conversations.
(b) How can you know that a female human being is in puberty stage? Give six points.

**SECTION C (15 Marks)**
**(Answer Question ten (10))**

10. A health officer of Mkunduru village has been punishing every family without rubbish heap and good pit latrine in their homes. Explain six points to support the idea of the village officer.

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**FORM TWO MOCK II**

**DODOMA REGION**

**MARKING GUIDE**

**033** **BIOLOGY**

**TIME: 2:30 Hours** **MAY 2024**

**QUESTION 01.**

| i | ii | iii | iv | v | vi | vii | viii | ix | x |
|---|---|---|---|---|---|---|---|---|---|
| C | B | B | D | A | A | C | A | C | A |

**QUESTION 02.**

| LIST A | i | ii | iii | iv | v |
|---|---|---|---|---|---|
| LIST B | B | F | A | D | E |

**QUESTION 03.**

(a) Cell A is **ANIMAL CELL** *(01 Mark)* Cell B is **PLANT CELL** *(01 Mark)*

(b) **The differences between Cell A and Cell B**

| S/N | CELL A (ANIMAL CELL) | CELL B (PALNT CELL) |
|---|---|---|
| i | It has irregular shape | It has regular shape |
| ii | It has no cell wall | It has cell wall |
| iii | It has small temporary vacuole | It has large permanent vacuole |
| iv | It stores carbohydrates in form of Glycogen | It stores carbohydrates in form of starch |

*01Mark@ Any four points* ***(Total 04Marks)***

(c) **The similarities between Cell A and Cell B**

(i) Both have Nucleus.

(ii) Both have cell membrane.

(iii) Both have Cytoplasm.

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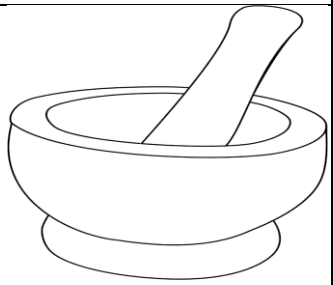
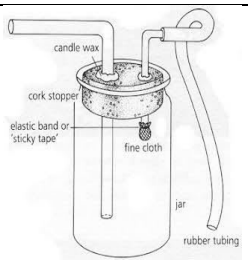
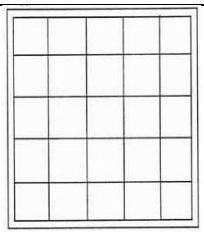
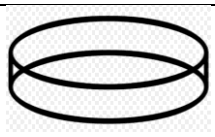
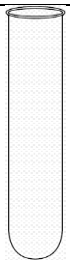
(iv) Both have Golgi bodies.

(v) Both have endoplasmic reticulum.

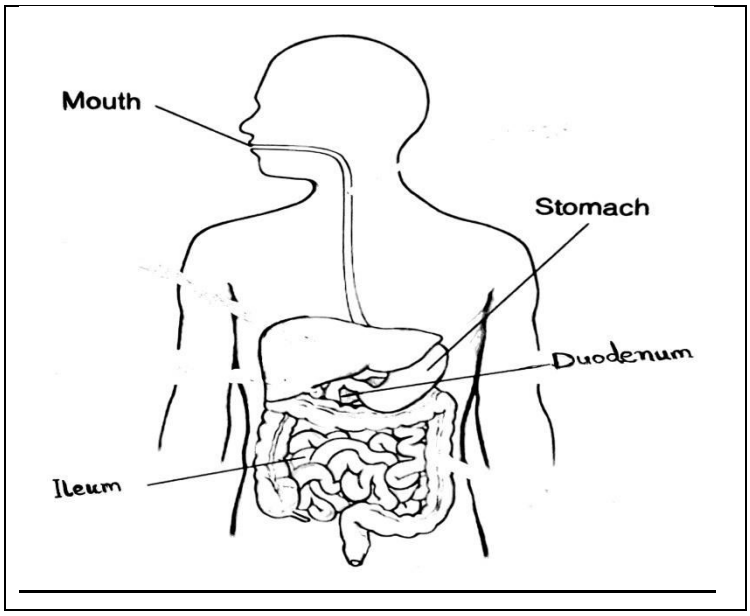
(vi) Both have Vacuoles.

*01Mark@ Any four points* ***(Total 04Marks)***

**QUESTION 04.**

| S/n | (a) | (b) | (c) | (d) | (e) |
|---|---|---|---|---|---|
| Apparatus Name | **Motor and Pestle** *(01Mark)* | **Pooter** *(01Mark)* | **Quadrat** *(01Mark)* | **Petri dish** *(01Mark)* | **Test tube** *(01Mark)* |
| Diagram |  *(01Mark)* |   *(01Mark)* |  *(01Mark)* |  *(01Mark)* |  *(01Mark)* |

**QUESTION 05.**




**The human digestive system showing only regions of digestions**

-Title ***(01Marks)*** -Box ***(01Marks)*** -Good diagram ***(02Marks)***

*-1.5 Marks@ Labeling (Mouth, Stomach, Duodenum and Ileum) (Total **06 Marks)***

***NB:*** Any student who will label all parts of the human digestive system instead of only parts where digestion ocuurs, will be given marks for only drawing(diagram).

**QUESTION 06.**

(a) Housefly *(02 Marks)*

(b) **The pathogens are:**

i. Vibrio cholerae ii. Salmonela typhi iii. Amoeba

*01Mark@ Any two pathogens of water borne diseases **(Total 02Marks)***

(c) **The diseases are:**

i. Cholera ii. Typhoid, iii. Dysentery

*01Mark@ Any two water borne diseases spread by Housefly **(Total 02Marks)***

(d) **The methods of preventing water-borne diseases are:**

i/ Drinking boiled or treated water.

ii/ Washing hands with soap and clean water before eating or after going toilet.

iii/ Relieving ourselves in a safe toilet.

iv/ Eating hot food all the time.

v/ Maintaining personal hygiene.

*01Mark@ Any four points* ***(Total 04Marks)***

**QUESTION 07.**

(a) -Artificial classification is the type of classification which is based on the observable features only***.(01 Mark)***

-Natural classification is the type of classification which is based on the observable and internal features by considering evolutionary trend or sequence. ***(01 Mark)***

(b) **The similarities between Fungi (Mushroom) and Plants (Moss plant):**

(i) Both Fungi and Plants have cells with cell walls.

(ii) Both Fungi and Plants are stationary organisms, they are unable to locomote,

(iii) Both are reproduce asexually by means of spores.

(iv) Both Fungi and Plants are anchored to soil for support and getting nutrients.

*02Mark@ Any four points* ***(Total 08Marks)***

**QUESTION 08.**

(a) **The functions of Microscope parts**

(i) **Mirror** reflects and directs light to the specimen under observation. ***(1.5Marks)***

(ii) -**Arm** Support the body tube, knobs, stage diaphragm and mirror.

**-Arm** is held when moving the microscope. ***(1.5Marks)***

(iii) **Base** provides firm support for the microscope ***(1.5Marks)***

(b) **SOLUTION**

Data given

-Eyepiece lens magnification = 10x

-Lower power objective lens magnification = 15x

-High power objective lens magnification = 40x

-Highest total magnification = Asked

From:

| **Highest total magnification = Eyepiece lens magnification X High power objective lens magnification** |
|---|

Highest total magnification = 10 X 40

= 400x

The Highest total magnification of the Microscope is 400x

**QUESTION 09.**

(a) **The principles of good manners during conversations are:**

(i) speaking in a clear voice. Do not shout or speak too soft.

(ii) Do not interrupting when other people are talking.

(iii) Do not using foul language.

(iv) Listening attentively to what the other person is saying.

(v) Avoiding dwelling on topics that are boring or embarrassing to other people.

(vi) Do not dominating the conversation. Allowing other people a chance to talk.

(vii)Being respectful when disagree with someone's opinions.

*01Mark@ Any four points* ***(Total 04Marks)***

(b) **The changes of Girls during puberty (Secondary sexual characteristics of Girls during puberty)**

(i) Breasts develop and increases in size.

(ii) Ovulation begin and menstruation start.

(iii) Hairs develop under the armpit and pubic areas.

(iv) Broadening of Hips and Pelvic girdles.

(v) Development of soft and high picked voice.

(vi) Increase in height and weight.

(vii) Changes in body ordour due the excessive production of Sweat and oil from glands.

(viii) Increase in secretion of mucus from the vagina.

(ix) Development of pimples in some Girls.

(x) Girls become more attracted to the members of opposite sex.

*01Mark@ Any six points* ***(Total 06Marks)***

**QUESTION 10.**

-**Any good introduction** About waste disposal ***(02 Marks)***

**The effects of poor wate disposal are**

(i) Wastes in streets and parks can be blown by wind and litter a wide area.

(ii) Wastes on River banks and beaches Pollutes aquatic environment.

(iii) Accumulated wastes act as breeding sites for vectors and pathogens hence they are source of infections.

(iv) Wastes harm wildlife and domestic animals, for example plastic bags choke or suffocate animals when swallowed.

(v) Accumulated wastes in water channels block drainage and leads to flooding during rainy seasons.

(vi) Rotten wastes such as vegetables and dead animals cause Air and water pollution.

(vii) Broken glasses in public places lead to accidents especially children when they use them as toys.

(viii) Wastes degrade the natural beaty of the environment.

(ix) Burning plastic and rubber wastes lead to air pollution and breathing difficulties.

(x) Sewage and clinical wastes transmit diseases.

(xi) Decomposing wastes emit unpleasant smell.

(xii) Plastic wastes prevent water from penetrating the soil as a results water accumulate outside the soil and act as breeding sites of vectors such as Mosquitoes and cockroaches.

*02Mark@ Any six six effects of poor waste disposal* ***(Total 12Marks)***

Any good conclusion **(01 Marks)**